ZentroMart Voice Support

A document-grounded customer-support agent for the VoiceOps 2026 capstone.

From customer speech to a grounded reply.

The system combines policy retrieval, verified synthetic order tools and conversation memory. Gnani provides both speech recognition and speech synthesis; support decisions remain in the shared text agent.

61.5%
RAG question pass rate
32/52 practice questions

77.1%
Conversation turn pass rate
74/96 customer turns

8/8
Voice batch completed
Saved end-to-end run

ONE TURN THROUGH THE SYSTEM

01 / LISTEN
Gnani STT
Customer audio to text

>

02 / REASON
Support agent
Retrieval, tools, context

>

03 / SPEAK
Gnani TTS
Reply text to speech

>

04 / RECORD
Saved artifacts
Text, audio and evidence

FOUR IMPLEMENTATION STAGES

1 **LLM baseline**
API-based assistant with role, uncertainty and action limits.

2 **Grounded retrieval**
Markdown sections, embeddings, FAISS/BM25 candidates and a multilingual cross-encoder reranker.

3 **Conversation + tools**
Order verification, contextual follow-ups, topic changes, ticket creation and human handover.

4 **Voice + batch**
Gnani STT/TTS, microphone conversation client and a folder-based evaluator with per-call artifacts.

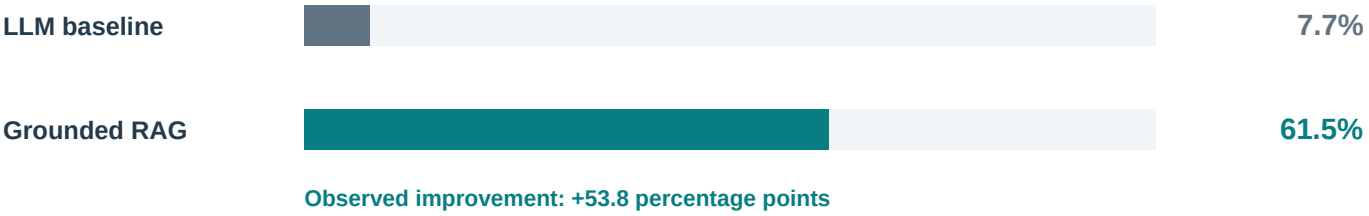
Dataset rules

Synthetic data only. Reference date: 20 October 2026. Business days: Monday-Saturday, excluding warehouse holidays; return windows use calendar days.

Measured performance

Latest saved practice runs. These results describe observed outputs, not an organizer-assigned score.

STEP 1 VS STEP 2 / QUESTION PASS RATE



Question metric	Step 1	Step 2
Questions passed	4/52	**32/52**
Decision accuracy	38.5%	**80.8%**
Required-fact accuracy	15.4%	**88.5%**
Grounded-answer accuracy	--	**65.4%**
Retrieval hit rate / MRR	--	**80.9% / 0.623**
Unsupported-question abstention	--	**60.0%**

MULTI-TURN CONVERSATIONS

74/96

Customer turns passed

77.1%

6/23

Complete conversations passed

26.1%

Control / behavior	Recorded accuracy
Tool calls / verification sequence	**99.0% / 100.0%**
Clarification / escalation	**75.0% / 100.0%**
Ticket creation / leakage-free rate	**83.3% / 100.0%**

Step 1: 24 Oct 2026, 18:06 IST | Step 2: 25 Oct 2026, 15:53 IST

Conversations: 25 Oct 2026, 17:23 IST

A conversation passes only when all its turns pass. Runs were recorded at different times; these results do not isolate the reranker's contribution or prove generalization to unseen cases.

Failure review & refinements

Observed practice failures guide the next changes. Automatically suggested root causes remain hypotheses.

20
RAG questions failed
Latest saved question run

9
Retrieval misses
Among answerable failures

6
Required-fact omissions
Categories may overlap

WHAT FAILED / WHAT TO IMPROVE

Failure pattern	Observed issue / likely cause	Planned correction
Multilingual retrieval	Hindi/Hinglish and multi-condition questions can miss the controlling section, including opened beauty-item rules.	**Inspect the candidate pool; improve language matching and include all relevant policy conditions.**
Incomplete answers	Some answers omit a specific deadline, fee, eligibility condition or refund destination.	**Use calculated policy facts and cover each requested detail with supporting evidence.**
Response decisions	Uncertainty, clarification and escalation wording can disagree with the expected decision.	**Review intent routing and the full response before changing rules; inspect evaluator false negatives.**
Conversation continuity	17 complete conversations fail at least one turn; tool correctness alone is insufficient.	**Recheck follow-ups, verification boundaries and handovers across complete conversations.**
Evaluation sensitivity	Decisions and facts use heuristics and text matching; wording and punctuation can affect scores.	**Normalize equivalent text consistently and review failures manually; preserve original agent outputs.**

RECENT IMPLEMENTATION REFINEMENTS

Order switching now clears private context and re-verifies the caller. Confirmed handovers export real ticket events. Deadlines use current policy and the holiday calendar. Evidence includes full chunk text and model-cited IDs; first replies disclose AI identity.

Voice delivery & next steps

End-to-end operation, reproducibility and the remaining evidence needed for a complete submission.

8
Files succeeded
Saved voice batch

0
Files failed
Batch processing errors

8
Audio replies present
Checked output WAV files

Batch summary: 8 succeeded, 0 failed.

VOICE BEHAVIOR & EVIDENCE

Gnani STT transcribes short customer turns; the same conversation agent generates a reply, and Gnani TTS saves the spoken output. Empty transcripts produce a spoken repeat request. Speech normalization removes visual formatting and groups eight-digit order IDs.

Saved per call	Purpose
transcript.txt / response.txt	**Recognized customer speech and the original agent reply.**
response.wav	**Gnani-generated spoken response.**
sources.json / trace.json	**Evidence text, section IDs, citations, actual tool calls and ticket details.**

Verification scope

Local regression checks exercise order routing, re-verification, tickets, calendar rules, source exports and unclear-audio recovery. A successful batch does not measure transcription accuracy, answer correctness or voice clarity.

Demo recording: 4:46 / maximum 2:00

A recording is present. Shorten it to two minutes or less while preserving the agent's actual answers. Keep a grounded question, a follow-up, and a handover or uncertainty response.

Before submission, review English, Hindi/Hinglish and unclear/noisy speech; publish the repository URL in README.md and refresh the evaluation after further changes. Keep credentials, virtual environments and the dataset copy out of the archive.

REPRODUCIBLE RUN COMMANDS

```
python run.py --input_folder input_folder --output_folder sample_outputs
python src/evaluation.py   |   python src/build_report.py
```